

Complete Portfolio Website Guide

defendwithgr.github.io

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Theme: Jekyll with Terminal Aesthetic

Table of Contents

1. [Project Overview](#)
 2. [What We Built](#)
 3. [Technical Architecture](#)
 4. [Educational: Core Concepts](#)
 5. [Maintenance Guide](#)
 6. [Troubleshooting](#)
 7. [Quick Reference](#)
-

Project Overview

Goal

Redesign your portfolio website (defendwithgr.github.io) from the al-folio academic theme to a modern, cybersecurity-focused terminal aesthetic similar to bhavsec.com.

What Was Accomplished

- Successfully migrated from al-folio to jekyll-theme-console
- Created custom terminal-style pages (whoami, projects, blog, resume)
- Implemented modern cyberpunk aesthetic with gradients and glows
- Added Azure home lab project to portfolio
- Updated AI project to reflect learning status
- Created auto-switching light/dark mode
- Deployed successfully to GitHub Pages

Key Features

- **Responsive Design:** Works on desktop and mobile
 - **Auto Dark Mode:** Switches based on system preference
 - **Terminal Aesthetic:** Monospace fonts, colorful prompts, cyberpunk styling
 - **Modern Enhancements:** Gradients, animations, card layouts, glowing effects
-

What We Built

File Structure

```
defendwithgr.github.io/
  _config.yml          # Main site configuration
  Gemfile              # Ruby dependencies
  _includes/
    head.html         # Custom CSS injection
  assets/
    css/
      custom.css      # Enhanced terminal styles
  index.md             # Home page
  whoami.md            # About page
  projects.md          # Projects listing
  blog.md              # Blog/writeups
  resume.md            # Resume/CV
  MAINTENANCE_GUIDE.md # Site maintenance reference
```

Pages Created

1. Home (index.md)

- Terminal-style introduction
- ASCII art welcome
- Quick links to main sections

2. whoami.md

- Professional background
- Current role as SOC Analyst
- Work experience summary
- Certifications (CCNA, ITILv4, ISC2 CC, Google Cybersecurity)
- Technical skills overview

3. projects.md

Contains 4 main projects: 1. **Cybersecurity Incident Response Plan** - Big Leaf Enterprise
2. **Vulnerability Assessment & Penetration Testing (VAPT)** - Active project 3. **Azure Home Lab** - Security operations environment (NEW) 4. **AI in Cybersecurity** - Learning journey (UPDATED)

4. blog.md

- TryHackMe writeups
- Medium articles
- Technical blog posts

5. resume.md

- Full professional experience

- Education background
 - Certifications
 - Technical skills categorized by:
 - Security Tools
 - SIEM & Analytics
 - Cloud Platforms
 - Programming Languages
-

Technical Architecture

Theme: `jeekyll-theme-console`

Why This Theme?

- **Minimalist design** perfect for terminal aesthetic
- **GitHub Pages compatible** via `remote_theme`
- **Lightweight and fast**
- **Easy to customize**

Configuration (`_config.yml`)

```
remote_theme: b2a3e8/jeekyll-theme-console
style: light
listen_for_clients_preferred_style: true
```

Custom Styling (`custom.css`)

Design System

Light Mode: - Background: Soft gradient (white to light gray) - Text: Dark gray (`#2b2d42`) - Accent colors: Cyan (`#00d9ff`), Neon Green (`#06ffa5`), Hot Pink (`#ff006e`)

Dark Mode (Cyberpunk): - Background: Deep dark (`#0a0e27`) - Text: Light gray (`#e0e6ed`) - Glowing effects: Multi-color radial gradients - Neon accents with text shadows

Key Visual Elements

1. **Terminal Prompts:** Green/cyan colored with glow effects
 2. **Headings:** Gradient underlines, shadow effects
 3. **Links:** Smooth transitions, hover effects
 4. **Cards:** Subtle shadows, gradient backgrounds
 5. **Animations:** Fade-in page entrance, blinking cursor
-

Educational: Core Concepts

What is Jekyll?

Jekyll is a static site generator that transforms plain text (Markdown files) into a complete website.

How It Works

Markdown Files (.md) → Jekyll Parser → HTML Website

↓
_config.yml
Gemfile
Liquid Templates

Benefits

- **Fast:** No database queries
- **Secure:** Static files, no server-side code
- **Free Hosting:** GitHub Pages support
- **Version Control:** Everything in Git
- **Easy Content:** Write in Markdown

Jekyll vs Static HTML

Feature	Jekyll	Static HTML
Content Updates	Edit .md files	Edit every HTML file
Reusable Components	Yes (includes)	Manual copy-paste
Blog Posts	Automatic listing	Manual updates
Themes	Easy to switch	Rebuild from scratch

What is a Gemfile?

A **Gemfile** lists all Ruby packages (gems) your Jekyll site needs.

Your Gemfile

```
source "https://rubygems.org"

gem "jekyll", "~> 4.4"           # Jekyll engine
gem "jekyll-theme-console"       # Terminal theme
gem "jekyll-feed"               # RSS feed
gem "jekyll-seo-tag"            # SEO optimization
gem "jekyll-sitemap"           # Sitemap generation
gem "jekyll-remote-theme"       # Remote theme support
```

Common Commands

```
bundle install    # Install all gems
bundle update     # Update gems to latest versions
bundle exec jekyll serve # Run development server
```

What is Liquid?

Liquid is Jekyll's templating language - it adds dynamic content to your pages.

Syntax Examples

Variables:

```
{{ site.title }}      # Outputs: "defendwithgr"
{{ page.title }}      # Outputs: current page title
```

Conditionals:

```
{% if page.title == "whoami" %}
  This is the about page!
{% endif %}
```

Loops:

```
{% for post in site.posts %}
  <li>{{ post.title }}</li>
{% endfor %}
```

Includes:

```
{% include head.html %}
```

Where You See Liquid

- Theme layout files (`_layouts/`)
- Include files (`_includes/head.html`)
- Custom navigation menus
- Dynamic content generation

Jekyll vs Hugo

Feature	Jekyll	Hugo
Language	Ruby	Go
Speed	Slower (Ruby interpreted)	Very fast (Go compiled)
Learning Curve	Easier for beginners	Steeper
GitHub Pages	Native support	Manual build required
Themes	Large ecosystem	Growing
Templating	Liquid (simple)	Go templates (complex)
Best For	Portfolios, blogs	Large documentation sites

Why We Chose Jekyll

1. **GitHub Pages Integration:** Zero configuration deployment
2. **Easier to Learn:** Simpler syntax and concepts
3. **Great Theme Ecosystem:** jekyll-theme-console works perfectly
4. **Lower Maintenance:** No build pipeline setup needed

GitHub Pages Deployment

How It Works

1. Push code to GitHub
↓
2. GitHub detects Jekyll site
↓
3. Builds site automatically
↓
4. Deploys to username.github.io

Configuration

- **Repository:** defendwithgr/defendwithgr.github.io
- **Branch:** terminal-redesign
- **URL:** <https://defendwithgr.github.io/>

Deployment Time

- Typical: 2-3 minutes after push
 - Check status: Settings → Pages or Actions tab
-

Maintenance Guide

Common Tasks

1. Update Content

Edit About Page File: whoami.md

```
## Who I Am  
[Edit your introduction here]
```

```
## Current Role  
[Update your position]
```

Add/Update Projects File: projects.md

```
## [5] New Project Name  
**Duration:** Start - End | **Type:** Category
```

Description of your project...

```
### Key Activities  
- Activity 1  
- Activity 2
```

```
**Technologies:** Tool1, Tool2, Tool3
```

Add Blog Posts File: blog.md

[2026-02-15] New TryHackMe Walkthrough
Brief description and [\[link to writeup\]\(url\)](#)

Update Resume File: resume.md - Add new experience under ## Professional Experience
- Add certifications under ## Certifications - Update skills under ## Technical Skills

2. Customize Visual Styling

Change Colors File: assets/css/custom.css

Light Mode:

```
body {  
    background: #ffffff;  
    color: #2b2d42;  
}  
  
.term-prompt {  
    color: #06ffa5; /* Change this */  
}  
  
.term-filename {  
    color: #ff006e; /* Change this */  
}
```

Dark Mode:

```
@media (prefers-color-scheme: dark) {  
    body {  
        background: #0a0e27;  
    }  
  
    .term-prompt {  
        color: #00ff00; /* Bright green */  
    }  
}
```

Popular Color Schemes Matrix Green:

```
.term-prompt { color: #00ff00; }  
.term-filename { color: #33ff33; }
```

Cyberpunk Pink/Cyan:

```
.term-prompt { color: #00d9ff; }  
.term-filename { color: #ff006e; }
```

Hacker Blue:

```
.term-prompt { color: #0066ff; }  
.term-filename { color: #00ccff; }
```

3. Add New Pages

Step 1: Create File Create `contact.md`:

```
---  
layout: page  
title: contact  
permalink: /contact.html  
---
```

```
# <span class="term-prompt">></span> contact <span class="term-cursor"> </span>
```

Your content here...

Step 2: Add to Navigation Edit `_config.yml`:

```
header_pages:  
  - index.md  
  - whoami.md  
  - projects.md  
  - blog.md  
  - resume.md  
  - contact.md # Add new page
```

4. Site Configuration

File: `_config.yml`

```
# Basic info  
title: Your Name  
email: your@email.com  
description: Your professional tagline  
  
# Footer  
footer: >  
  © 2026 Your Name |  
  <a href="https://linkedin.com/in/yourprofile">LinkedIn</a>  
  
# Theme  
remote_theme: b2a3e8/jekyll-theme-console  
style: light # or dark  
listen_for_clients_preferred_style: true
```

Workflow

Local Development

Setup (One-time)

```
# Install dependencies  
bundle install  
  
# Start development server  
bundle exec jekyll serve
```

Development Loop

```
# 1. Make changes to .md files or CSS  
nano whoami.md  
  
# 2. View at http://localhost:4000  
# (auto-refreshes on file save)  
  
# 3. Commit when satisfied  
git add -A  
git commit -m "Updated whoami page"  
git push origin terminal-redesign
```

Deployment Workflow

Quick Updates (No Local Testing)

```
# Edit files  
nano projects.md  
  
# Commit and push  
git add projects.md  
git commit -m "Added new project"  
git push origin terminal-redesign  
  
# Wait 2-3 minutes  
# Check https://defendwithgr.github.io/
```

Full Testing Workflow

```
# 1. Edit files  
nano resume.md  
  
# 2. Test locally  
bundle exec jekyll serve  
  
# 3. View at http://localhost:4000
```

```
# 4. Deploy
git add -A
git commit -m "Updated resume"
git push origin terminal-redesign

# 5. Verify deployment
# GitHub Actions: Check for green checkmark
# Live site: https://defendwithgr.github.io/
```

Troubleshooting

Common Issues

1. Site Not Updating on GitHub Pages

Symptoms: Changes pushed but old site still showing

Solutions:

```
# Solution 1: Hard refresh browser
# Mac: Cmd + Shift + R
# Windows: Ctrl + Shift + R

# Solution 2: Check GitHub Actions
# Go to: github.com/defendwithgr/defendwithgr.github.io/actions
# Look for green checkmark on latest run

# Solution 3: Wait longer
# GitHub Pages can take 3-5 minutes sometimes

# Solution 4: Clear browser cache
# Settings → Privacy → Clear browsing data
```

2. Local Build Fails

Symptoms: `bundle exec jekyll serve` errors

Solutions:

```
# Solution 1: Reinstall dependencies
bundle install

# Solution 2: Clear cache
bundle exec jekyll clean
bundle exec jekyll serve

# Solution 3: Update gems
bundle update
```

```
# Solution 4: Check for syntax errors in _config.yml  
# Use a YAML validator online
```

3. CSS Changes Not Showing

Symptoms: Updated custom.css but no visual changes

Solutions:

```
# Solution 1: Hard refresh browser  
# Cmd+Shift+R (Mac) or Ctrl+Shift+R (Windows)  
  
# Solution 2: Verify file saved  
ls -la assets/css/custom.css  
  
# Solution 3: Check _includes/head.html exists  
cat _includes/head.html  
  
# Solution 4: Clear Jekyll cache  
bundle exec jekyll clean  
  
# Solution 5: Verify CSS linking in head.html  
# Should have:  
# <link rel="stylesheet" href="{ '/assets/css/custom.css' | relative_url }">
```

4. GitHub Pages Build Failing

Symptoms: Red X in Actions tab

Solutions:

```
# Check the error logs:  
# 1. Go to github.com/dependwithgr/dependwithgr.github.io/actions  
# 2. Click on failed run  
# 3. Read error message  
  
# Common fixes:  
# - Use remote_theme instead of theme:  
remote_theme: b2a3e8/jekyll-theme-console  
  
# - Remove unsupported plugins from _config.yml  
# - Check for YAML syntax errors  
# - Ensure Gemfile has jekyll-remote-theme
```

5. Content Not Displaying

Symptoms: Blank pages or missing content

Solutions:

```
# Check front matter in .md files:  
---
```

```
layout: page
title: whoami
permalink: /whoami.html
---
```

```
# Ensure file is in root directory
# Verify header_pages in _config.yml includes the file
```

Quick Reference

File → Purpose Mapping

File	What It Does
_config.yml	Site settings, navigation, theme configuration
Gemfile	Ruby packages and dependencies
index.md	Home page content
whoami.md	About/profile page
projects.md	Projects listing
blog.md	Blog/writeups index
resume.md	Full resume/CV
assets/css/custom.css	Custom styling (colors, fonts, effects)
_includes/head.html	Loads custom CSS into pages
MAINTENANCE_GUIDE.md	This guide!

Essential Commands

Local Development

```
# Start dev server
bundle exec jekyll serve

# Open in browser
open http://localhost:4000

# Stop server
Ctrl+C
```

Git/Deployment

```
# Quick commit and deploy
git add -A
git commit -m "Update content"
git push origin terminal-redesign

# Check status
git status
```

```
# View commit history
git log --oneline

# Create new branch
git checkout -b new-feature
```

Troubleshooting

```
# Reinstall gems
bundle install

# Clear Jekyll cache
bundle exec jekyll clean

# Check Ruby/Jekyll versions
ruby -v
jekyll -v
bundle -v
```

Color Reference

Current Color Palette

Light Mode: - Background: #f8f9fa → #e9ecef (gradient) - Text: #2b2d42 - Prompt: #06ffa5 (neon green) - Path: #00d9ff (cyan) - Filename: #ff006e (hot pink) - Links: #0066ff → #00d9ff (hover)

Dark Mode: - Background: #0a0e27 - Text: #e0e6ed - Prompt: #00ff00 (bright green, glowing) - Path: #00d9ff (cyan, glowing) - Filename: #ff6b6b (bright red, glowing) - Links: #00d9ff → #06ffa5 (hover)

Terminal Color Classes

```
<span class="term-prompt">></span>    <!-- Green prompt -->
<span class="term-path">~</span>      <!-- Cyan/green path -->
<span class="term-filename">file.md</span> <!-- Red filename -->
<span class="term-cursor"></span>      <!-- Blinking cursor -->
```

Useful Links

Documentation

- Jekyll Docs: <https://jekyllrb.com/docs/>
- Liquid Syntax: <https://shopify.github.io/liquid/>
- Markdown Guide: <https://www.markdownguide.org/>
- Theme Docs: <https://github.com/b2a3e8/jekyll-theme-console>

Tools

- YAML Validator: <https://www.yamllint.com/>

- Markdown Preview: <https://markdownlivepreview.com/>
- Color Picker: <https://colorhunt.co/>

Your Site

- Live Site: <https://defendwithgr.github.io/>
 - Repository: <https://github.com/defendwithgr/defendwithgr.github.io>
 - GitHub Actions: <https://github.com/defendwithgr/defendwithgr.github.io/actions>
 - Settings: <https://github.com/defendwithgr/defendwithgr.github.io/settings/pages>
-

Key Learnings Summary

Technical Skills Gained

1. **Static Site Generators**
 - Understanding Jekyll architecture
 - Working with Gemfiles and Ruby gems
 - Using Liquid templating language
2. **GitHub Pages**
 - Deployment workflows
 - Branch management
 - Actions/CI-CD basics
3. **CSS Styling**
 - Custom theming
 - Responsive design
 - Dark mode implementation with media queries
 - CSS animations and transitions
4. **Version Control**
 - Git branching strategy
 - Commit best practices
 - Repository management

Design Principles Applied

1. **Visual Hierarchy**
 - Clear headings with underlines
 - Card-based layouts for sections
 - Proper spacing and margins
2. **Color Theory**
 - Neon accents for terminal aesthetic
 - High contrast for readability
 - Complementary color schemes (cyan/pink/green)
3. **User Experience**
 - Fast load times (static site)
 - Responsive mobile design
 - Auto dark/light mode
 - Smooth animations

4. Accessibility

- High contrast text
 - Readable fonts (monospace but clear)
 - Semantic HTML structure
-

Project Timeline

Phase 1: Planning & Research

- Explored bhavsec.com for inspiration
- Researched Jekyll themes
- Selected jekyll-theme-console

Phase 2: Theme Setup

- Removed al-folio theme
- Installed jekyll-theme-console
- Configured `_config.yml`
- Set up `remote_theme` for GitHub Pages compatibility

Phase 3: Content Migration

- Created terminal-style pages (index, whoami, projects, blog, resume)
- Migrated content from al-folio
- Added terminal prompts and styling

Phase 4: Visual Enhancement

- Initial CSS with purple glow background
- Fixed readability issues (text colors)
- Switched to light theme with dark mode support
- Major CSS enhancement with cyberpunk aesthetics

Phase 5: Content Updates

- Added Azure Home Lab project
- Updated AI project to learning status
- Refined project descriptions

Phase 6: Final Polish

- Enhanced CSS with gradients and animations
 - Added card layouts
 - Implemented smooth transitions
 - Created comprehensive documentation
-

Final Notes

What Makes This Portfolio Special

1. **Unique Terminal Aesthetic:** Stands out from typical portfolio sites
2. **Cybersecurity Focus:** Design matches your field (hacker/terminal theme)
3. **Modern & Professional:** Balance between creative and professional
4. **Easy to Maintain:** Simple Markdown files, clear structure
5. **Auto Dark Mode:** Respects user preferences
6. **Fast & Secure:** Static site, no vulnerabilities

Future Enhancement Ideas

Content

- ☐ Add detailed project pages with screenshots
- ☐ Create blog posts for TryHackMe walkthroughs
- ☐ Add vulnerability research writeups
- ☐ Include CTF achievements

Features

- ☐ Add a contact form
- ☐ Implement search functionality
- ☐ Add tags/categories for projects
- ☐ Include project demos/videos

Design

- ☐ Add subtle particle effects background
- ☐ Create custom 404 page
- ☐ Add loading animations
- ☐ Include Easter eggs for fun

Remember

Content is King: Focus on showcasing your work and skills

Keep It Updated: Regular updates show active engagement

Get Feedback: Share with peers and mentors

Monitor Analytics: See what people view most

Stay Authentic: Let your personality show through

Good luck with your portfolio!

If you need any help, refer back to this guide or the `MAINTENANCE_GUIDE.md` file in your repository.

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